

Electrical Installation Practice 2

LSEGG304A
9080D

Protection Devices

Fuses

Content

- Operating principles of fuses
- Inverse time/current characteristic of fuses
- Fuses used as fault current limiters
- Fuses and fault loop impedance
- Fuse selection

Circuit Protection

What is it's Job?

- Protect circuit wiring against overheating & deterioration due to overloads
- Quickly interrupt a short circuit so:
 - Limit the energy let through
 - Reduce the touch potentials rising too high

Circuit Protection

- **Fuses**
- **Circuit breakers**

Next lesson

Fuses

- Two Basic types
 - Rewireable

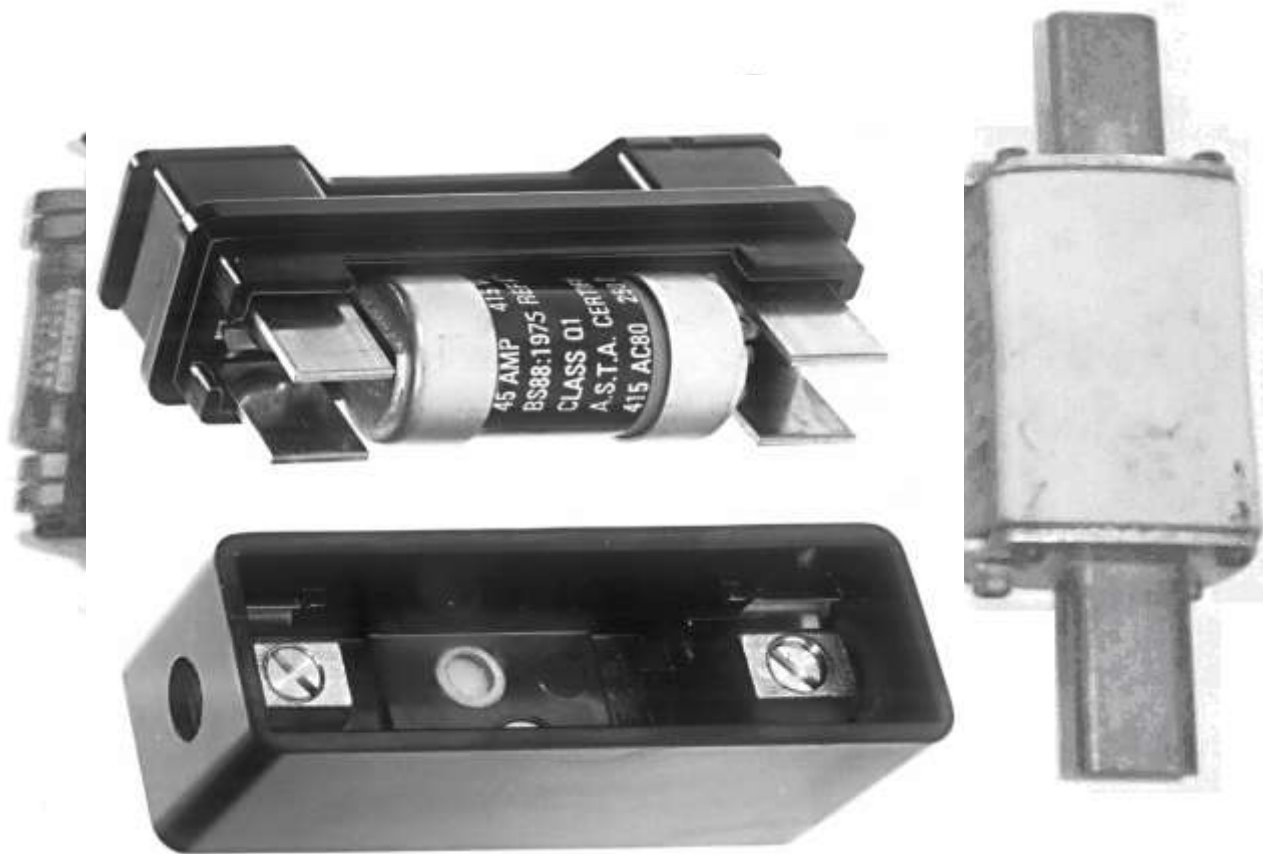


Fuses

- Two Basic types
 - Rewireable
 - Sealed
 - Gas filled Glass automotive fuses
 - Silica filled HRC

HRC Fuses

(High Rupture Capacity)

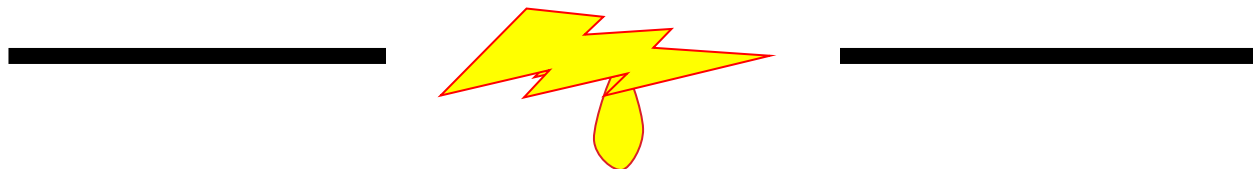


Fuses

- Two Basic types
 - Rewireable
 - Sealed
 - Gas filled Glass automotive fuses
 - Silica filled HRC
- Still provides the greatest fault handling capacity for the size

How Does A Fuse Work?

- As current increases, fuse element heats up
- Becomes liquid, and falls away.
- An arc forms between the ends.
- The ends melt away, the arc becomes longer.
- Eventually the gap is so great that it is too long for the arc. Current stops flowing

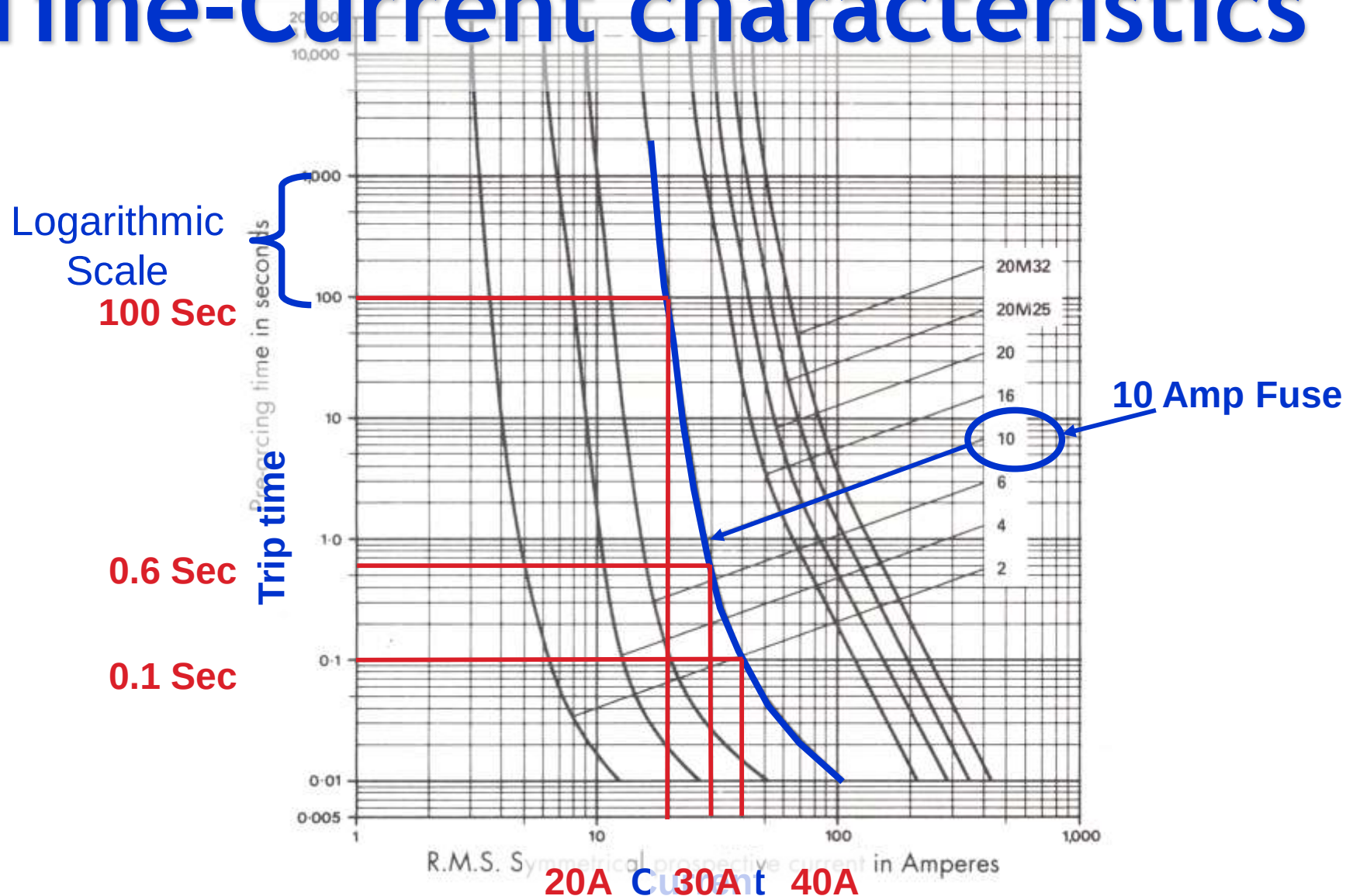


$$H = I^2t$$

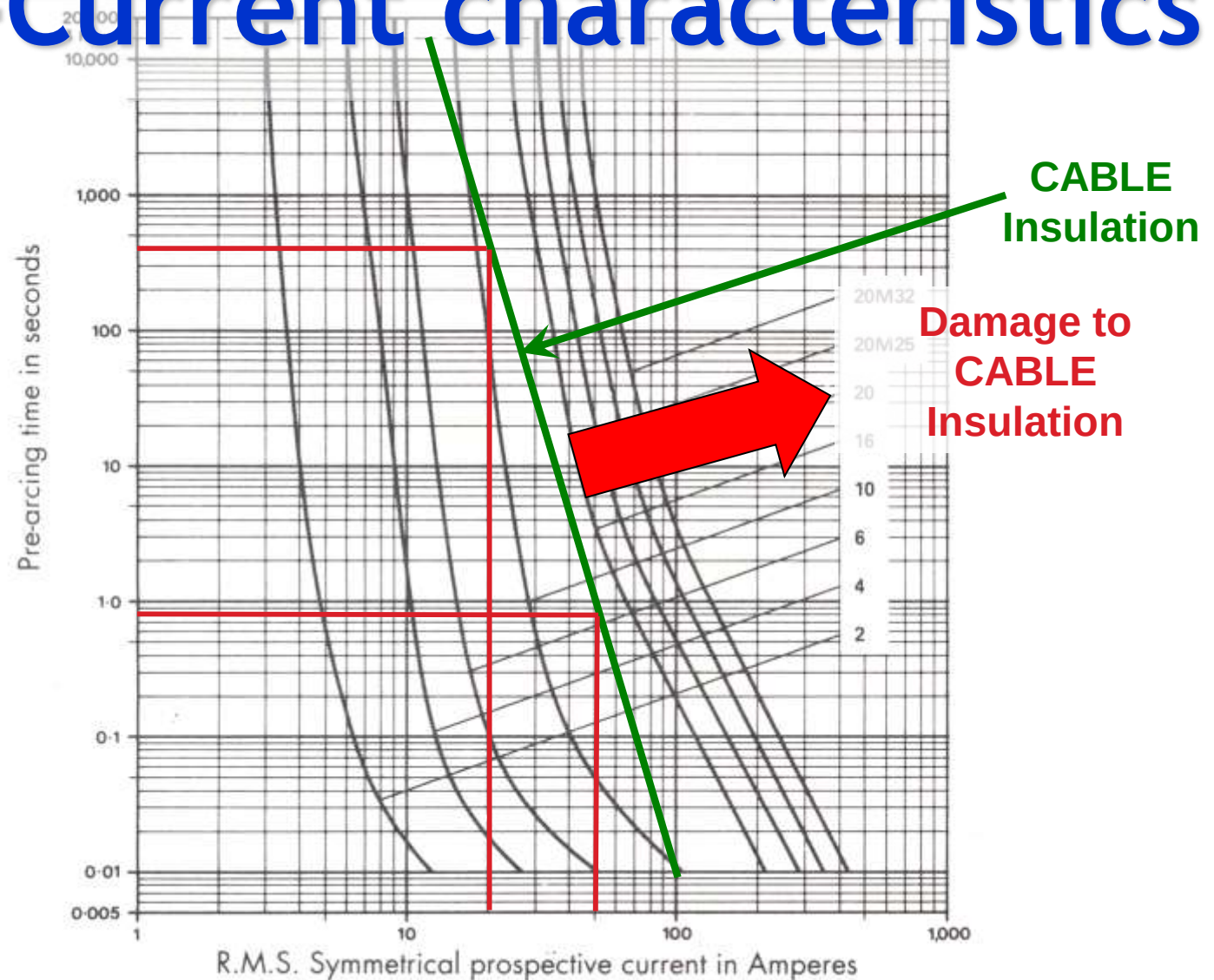
Definitions

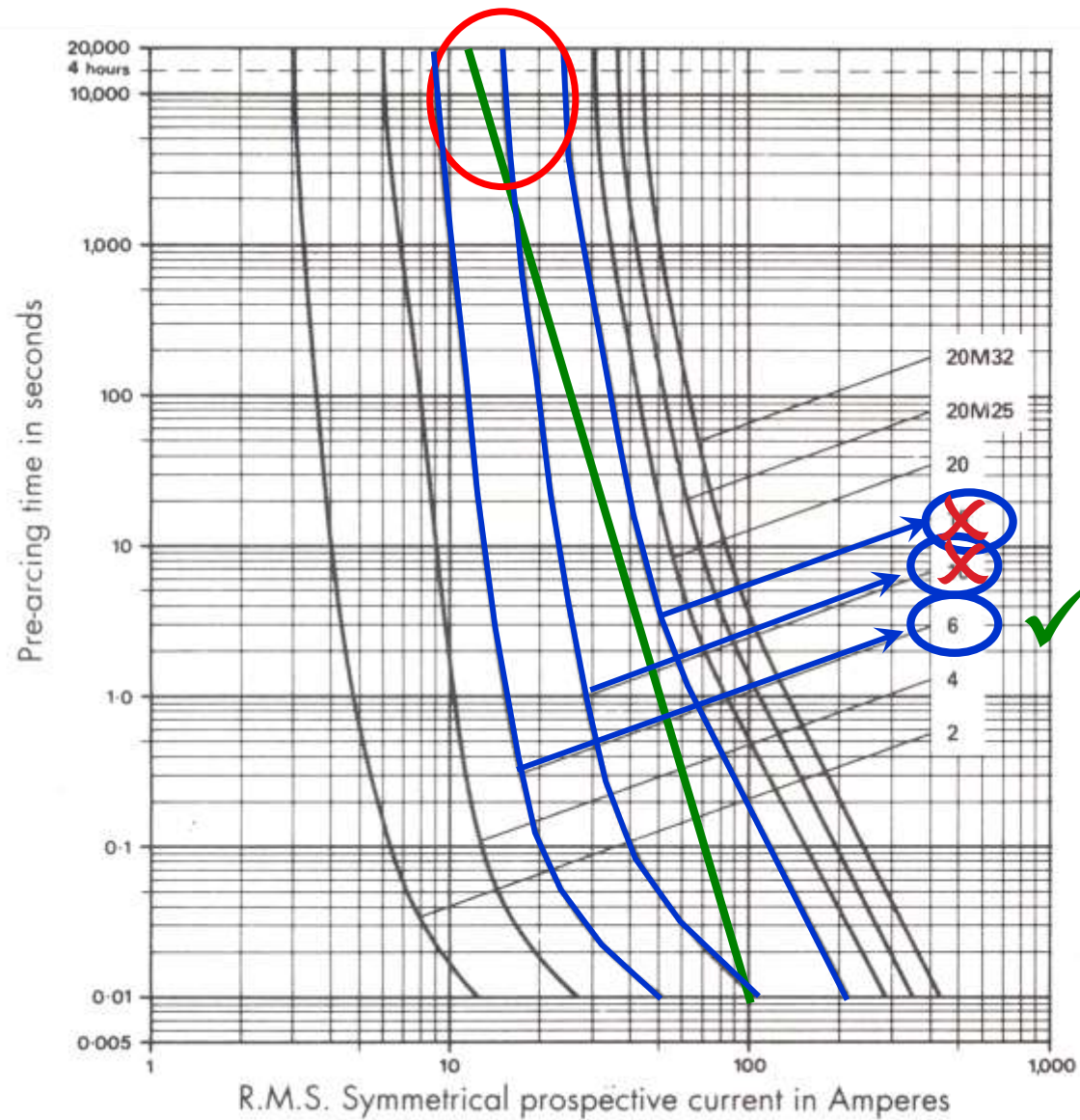
- Current Rating The maximum current that a fuse can carry continuously without deterioration
- Voltage Rating The maximum voltage that the fuse can safely operate.
- Time-Current characteristics

Time-Current characteristics



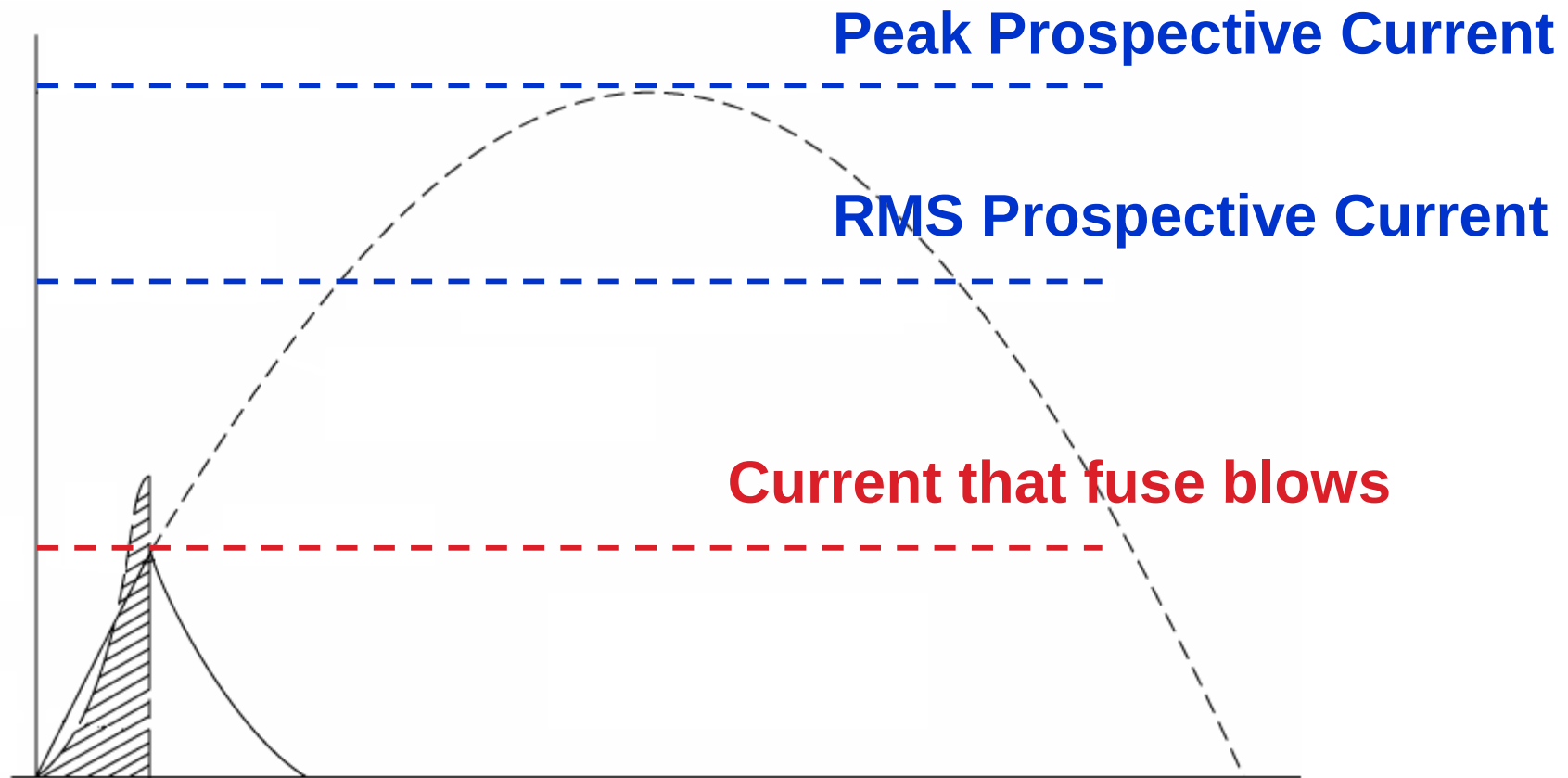
Time-Current characteristics

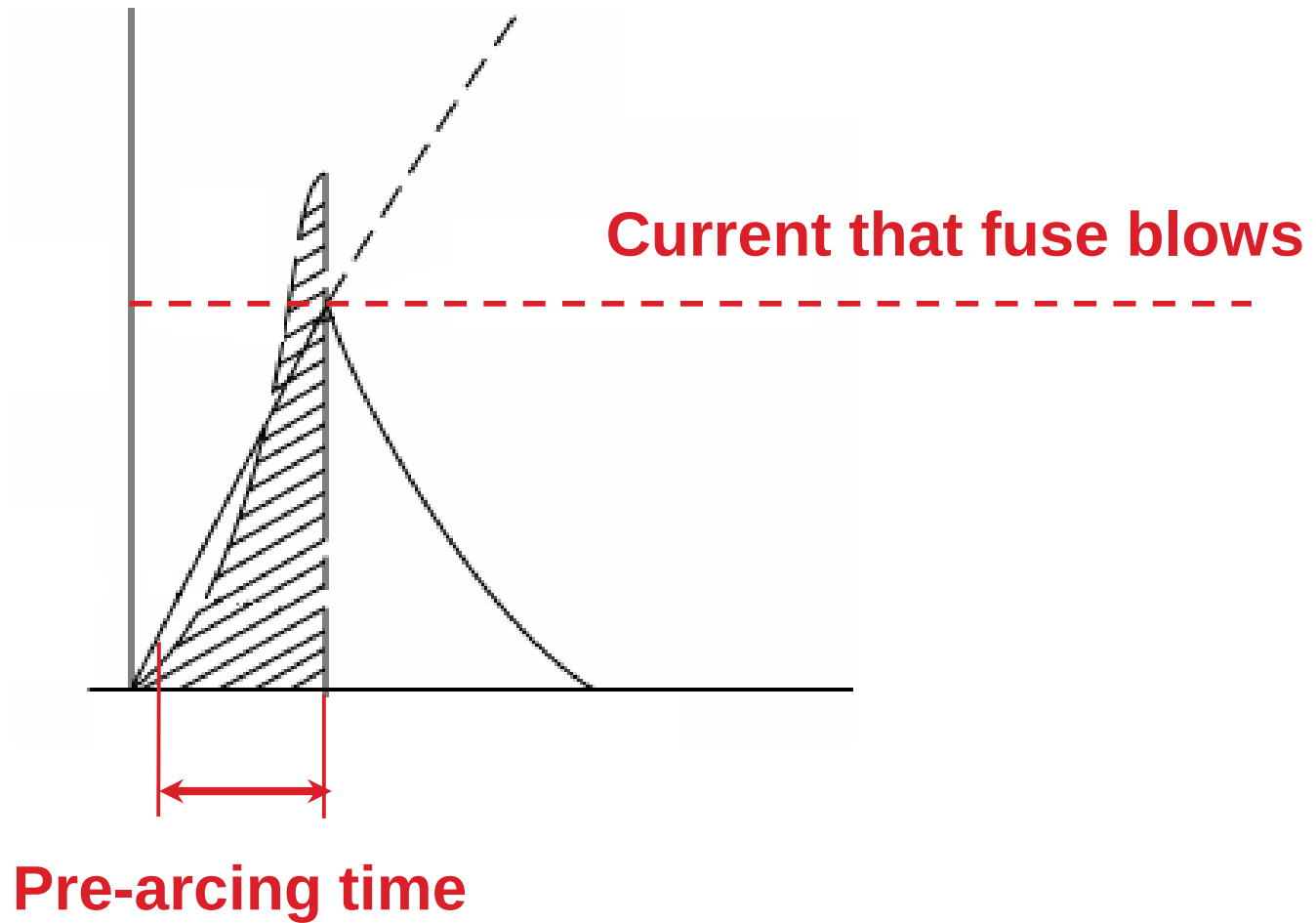




Definitions

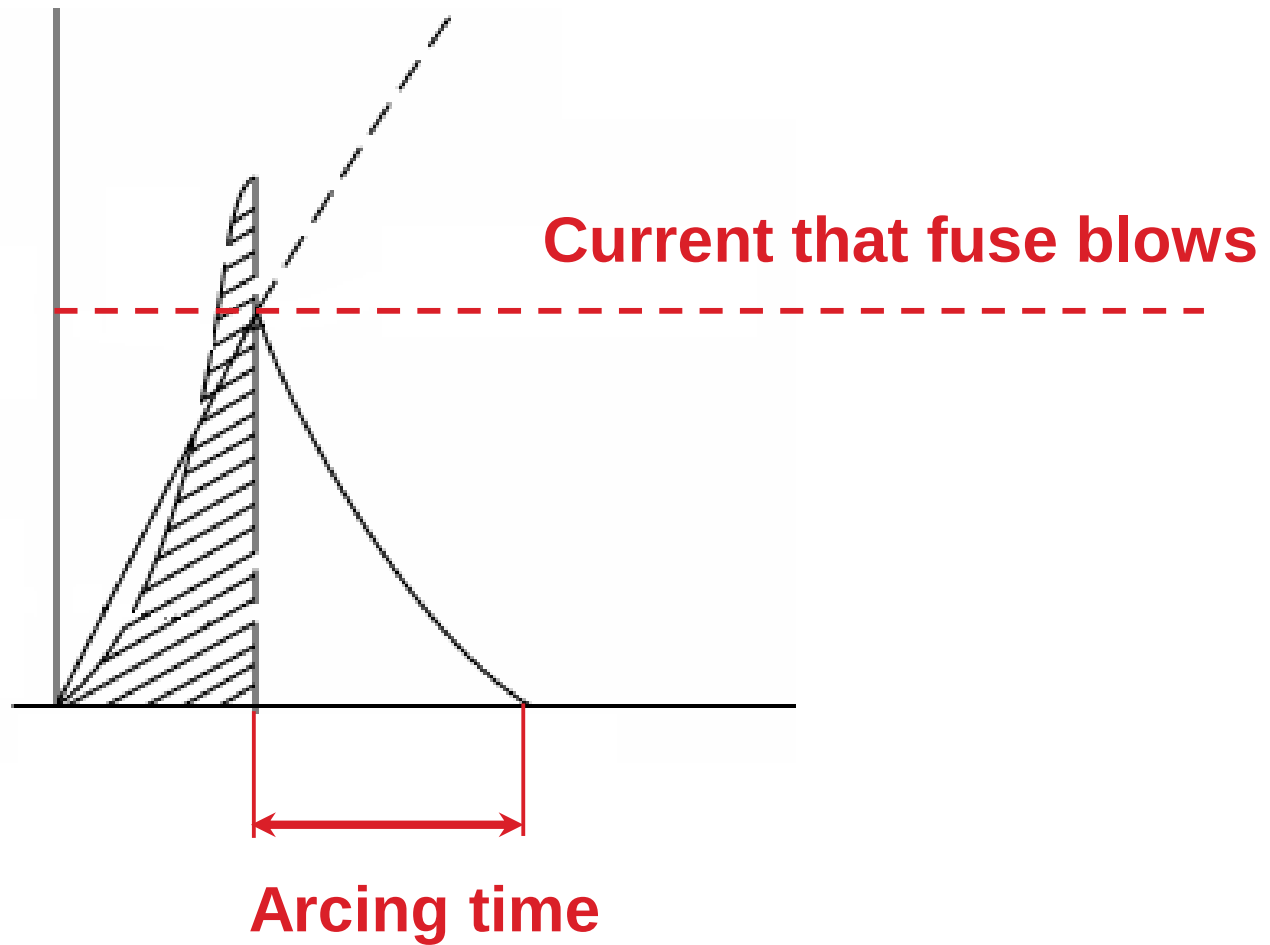
- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time





Definitions

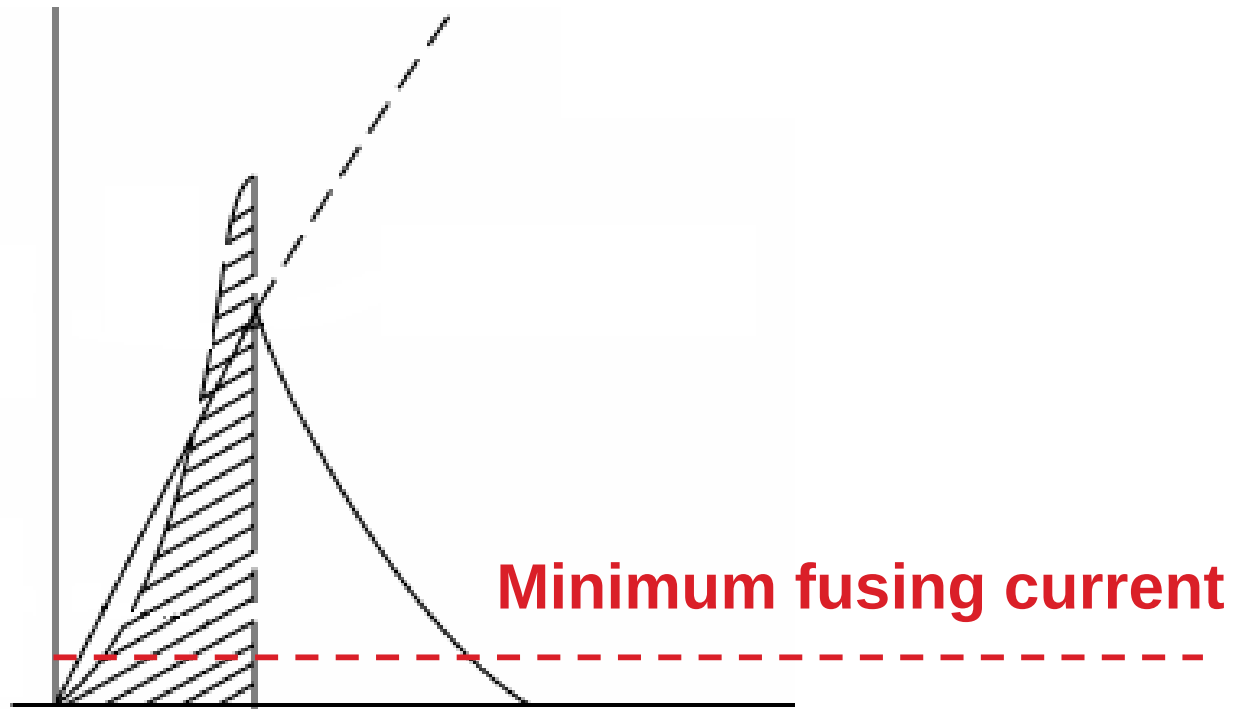
- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time
- Arcing time



Definitions

- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time
- Arcing time
- Minimum fusing current

The minimum current that the fuse element will start to melt



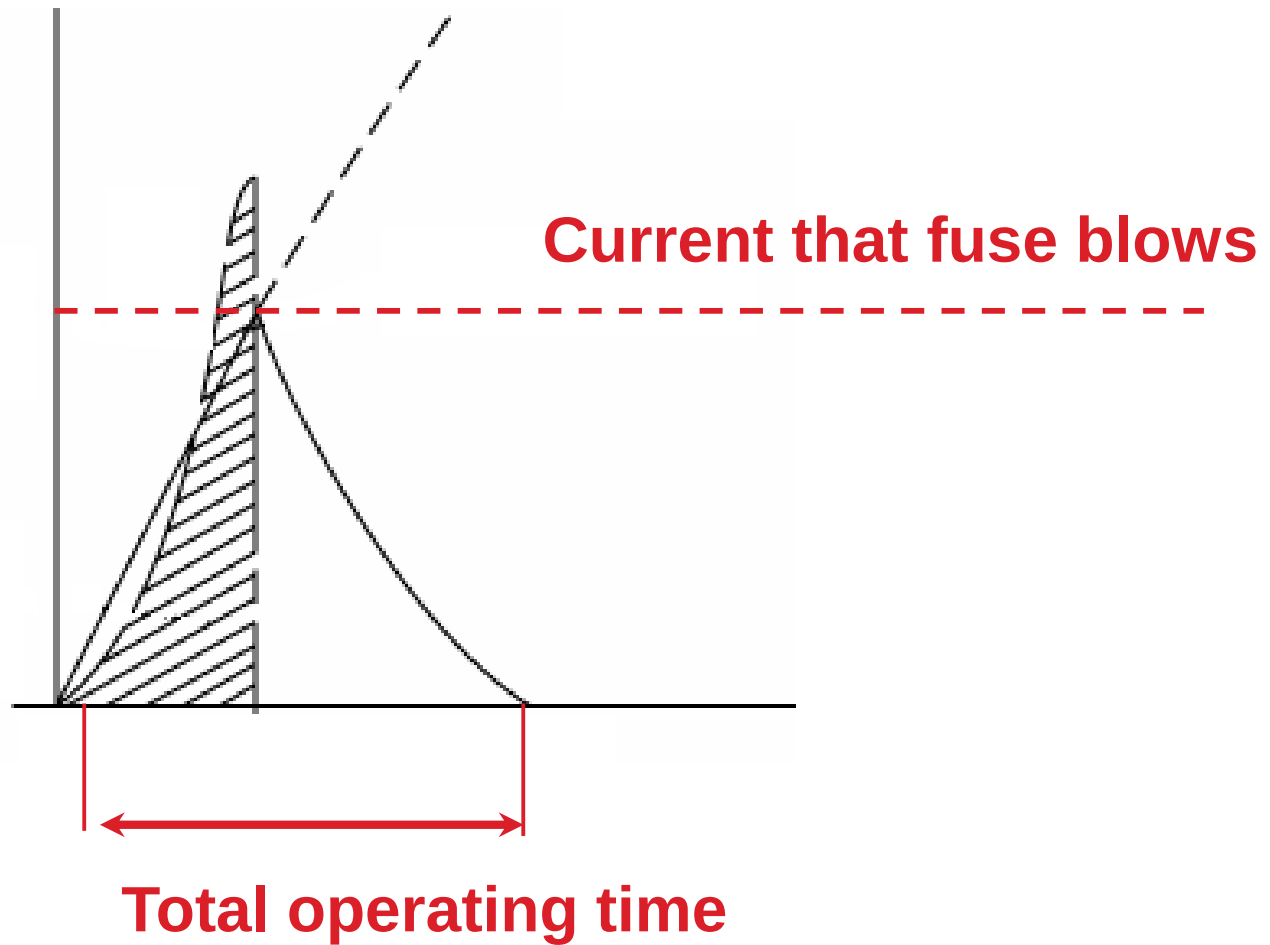
Definitions

- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time
- Arcing time
- Minimum fusing current
- Fusing Factor = $\frac{\text{Min fusing current}}{\text{Current rating}}$

Typical values are in the order of 1.5 to 2

Definitions

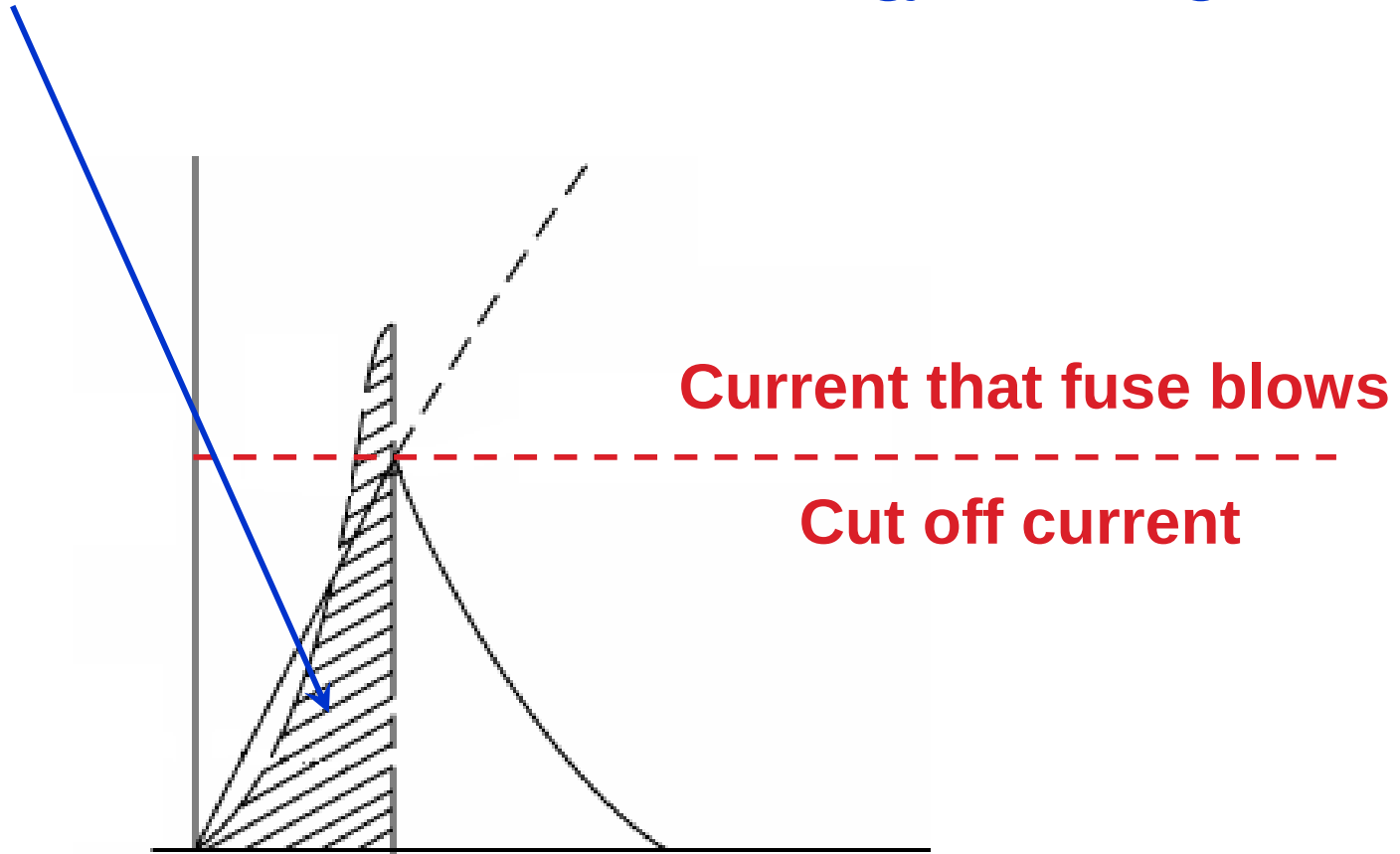
- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time
- Arcing time
- Minimum fusing current
- Fusing Factor
- Total operating time



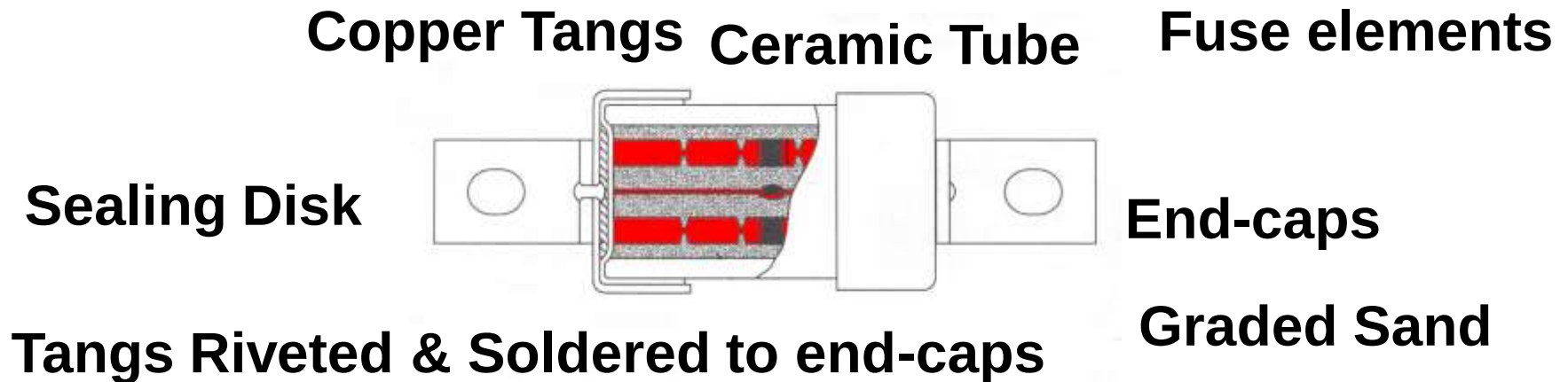
Definitions

- Current Rating
- Voltage Rating
- Time-Current characteristics
- Pre-arcing time
- Arcing time
- Minimum fusing current
- Fusing Factor
- Total operating time
- Cut-off current

Shaded area = $I^2 t$ OR Energy let through



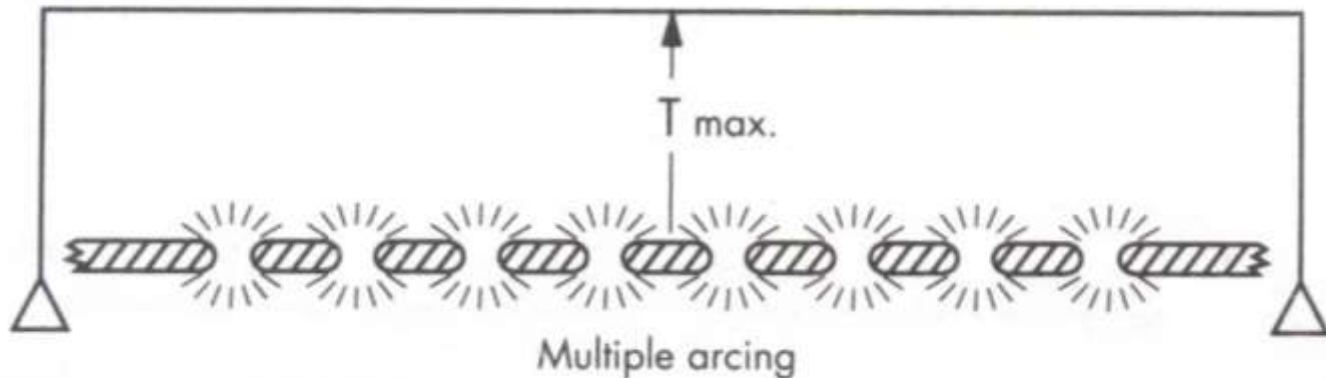
Construction



With multiple arc points the time to blow is faster



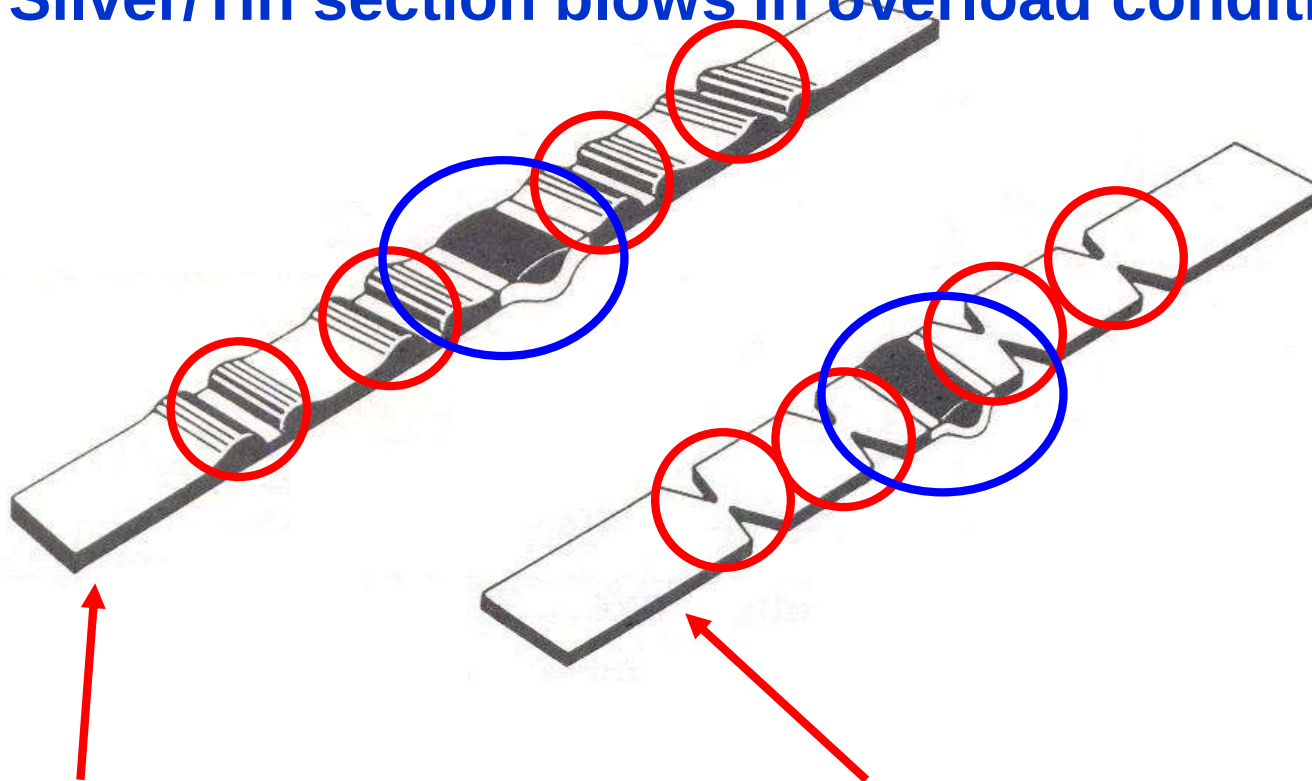
Standard HRC Fuse Element



Fusible Elements

Copper sections that blow in short circuit conditions

Silver/Tin section blows in overload conditions



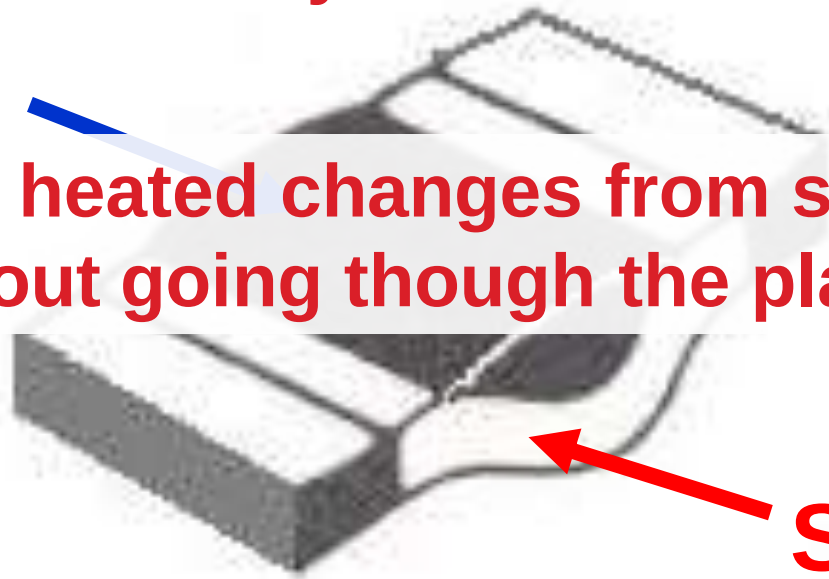
Silverbond Rolled Element

Standard Element

With overload currents the tin & silver combine to produce an alloy that melts at 230°C not at Silver's melting point of 960°C **Eutectic Alloy**

Tin

When heated changes from solid to liquid without going through the plastic region



Silver

Also known as the “M” effect

HRC HV Fuse Types

- **Distribution/Transformer**

- Transformer inrush currents
(high current for short period of time)
- Overload protection
- Operate in reasonable period of time with regard to secondary short circuit

Motor circuit

- Fast operation for short circuits
- High inrush for long period of time

HRC Low Voltage

- High breaking capacity & energy limitation.
- Restriction of electro-mechanical stress on cables and busbars
- Reliable short circuit and back-up protection.
- Accurate discrimination.
- Low over-current protection.
- Non-deterioration due to no moving parts

HRC Semiconductor Fuses

- Electronics and power cables
- Energy let through
- Must be very fast
- Fusing element
- than standard



motors or

s.

tion

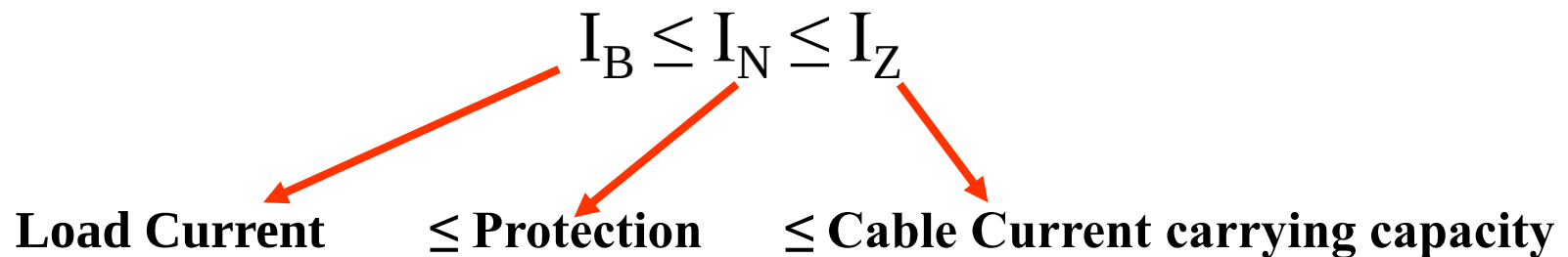
r, & thinner

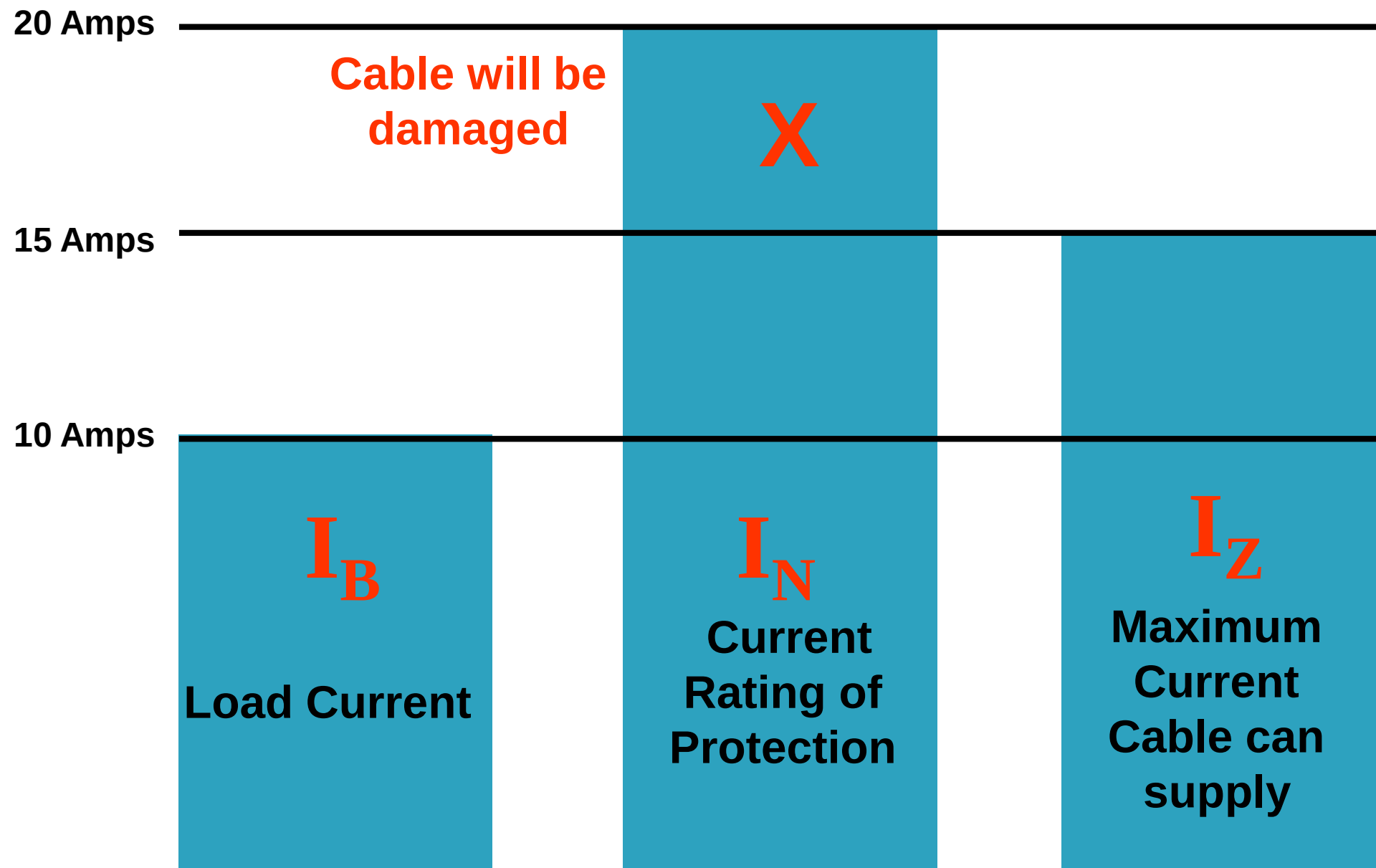
Matching Protection to a Cable

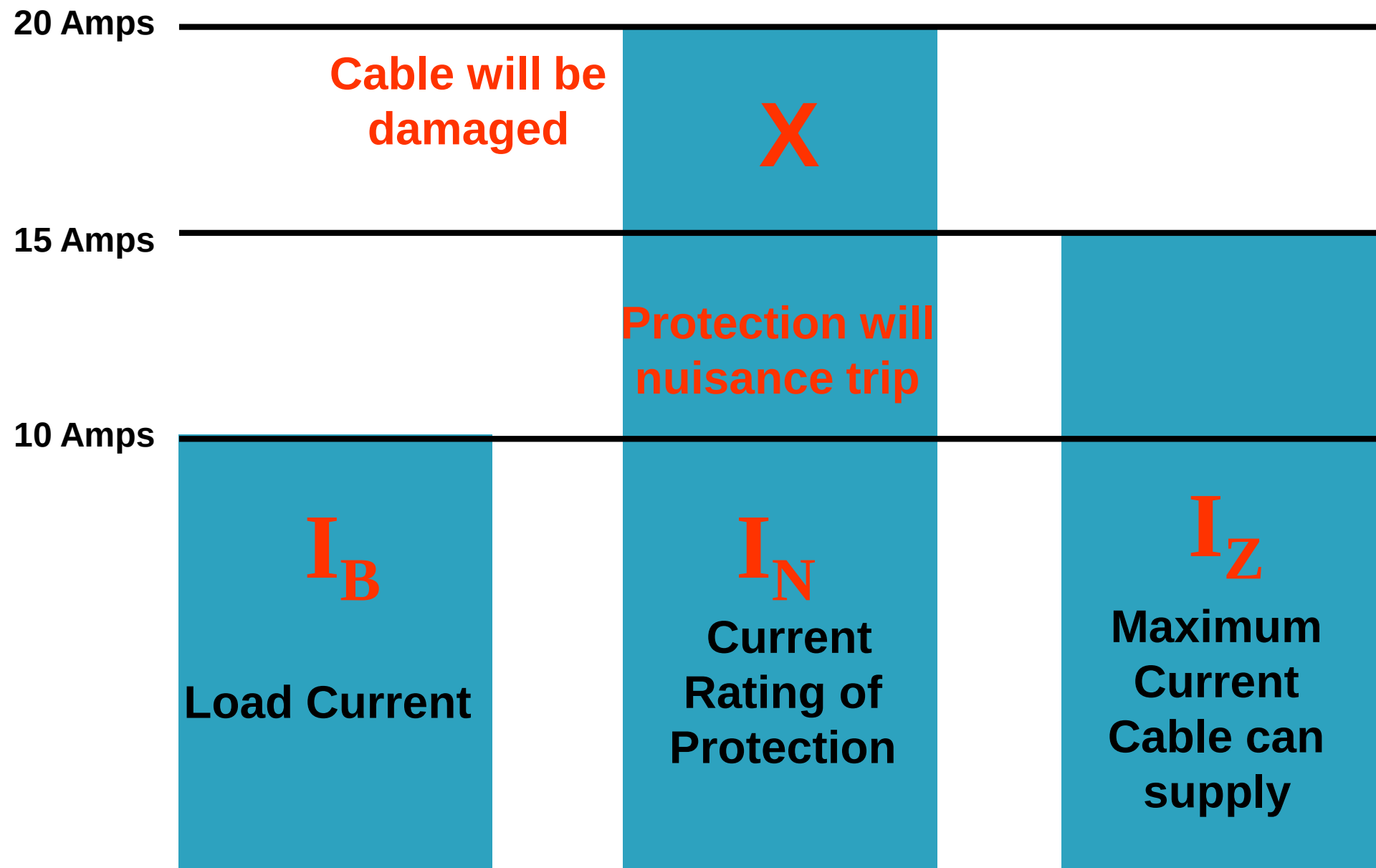
A cable's current carrying capacity must be equal or larger than the load current

Circuit protection must be equal to or smaller than the cable's current carrying capacity

**AS/NZS 3000:2007
Clause 2.5.3.1
Page 76**





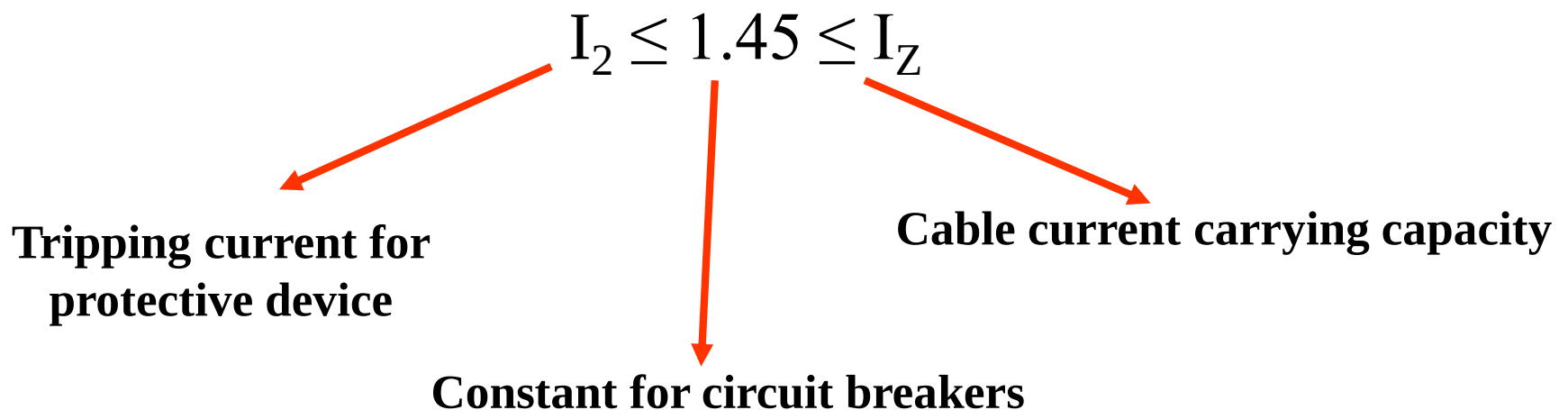


Matching Protection to a Cable

But the protection must match the cable

AS/NZS 3000:2007
Clause 2.5.3.1
Page 76

For circuit breakers

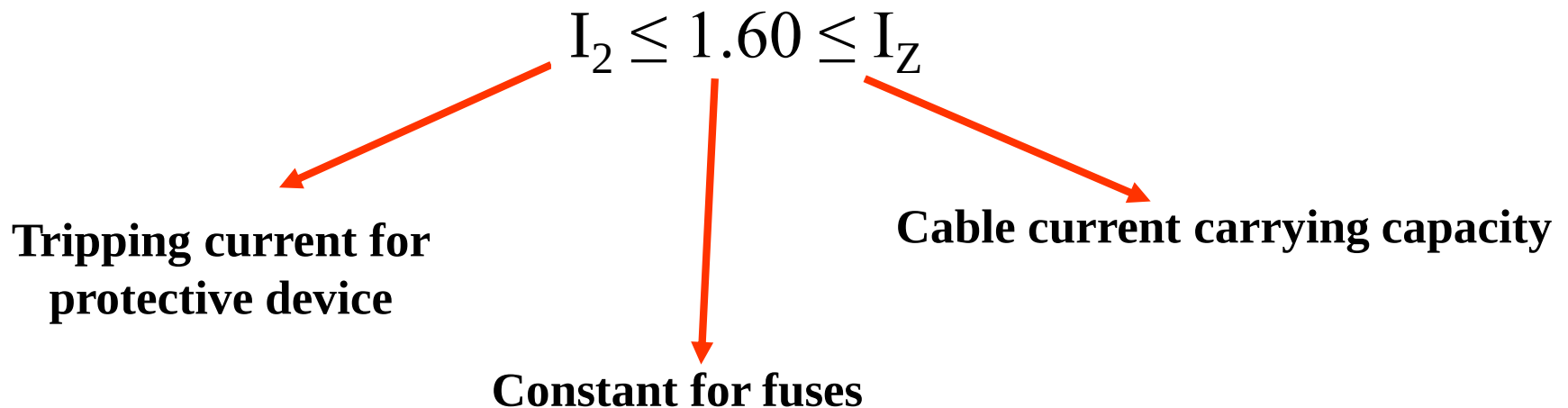


Matching Protection to a Cable

But the protection must match the cable

AS/NZS 3000:2007
Clause 2.5.3.1
Page 76

For Fuses



However

A cable can withstand a overload current of 1.45 x it's rating before the insulation is damaged

14.5 Amps

14.5 Amps

10 Amps

$$1.45 \times 10A =$$

I_z

Maximum
Current
Cable can
supply

Circuit Breakers

A Circuit breaker's trip curve is matched to a cable's curve

Tripping current is $1.45 \times$ rated current

14.5 Amps

10 Amps

$$1.45 \times 10A = 14.5 \text{ Amps}$$

I_N

Current
Rating of
Protection

I_Z

Maximum
Current
Cable can
supply

Fuses

A Fuse's trip curve is different to a cable's curve

16 Amps

Tripping current is 1.6 x rated current

X

14.5 Amps

Cable will be
damaged

10 Amps

I_N

Current
Rating of
Protection

I_Z

Maximum
Current
Cable can
supply

16 Amps

$$1.6 \times 10A =$$

Fuses

16 Amps

14.5 Amps

10 Amps

1.6

9 Amps

$$\frac{14.5}{1.6} =$$

I_N

Current
Rating of
Protection

I_Z

Maximum
Current
Cable can
supply

Fuses

16 Amps

14.5 Amps

10 Amps

0.9



$$10 \times 0.9 = 9 \text{ Amps}$$

I_N

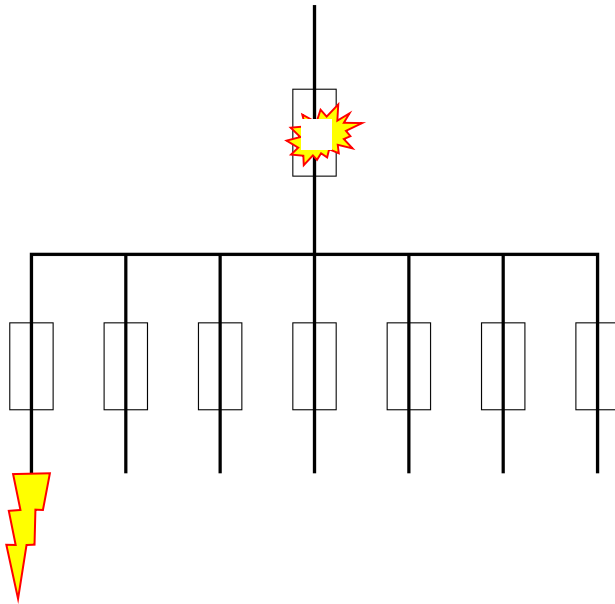
Current
Rating of
Protection

I_Z

Maximum
Current
Cable can
supply

Poor discrimination

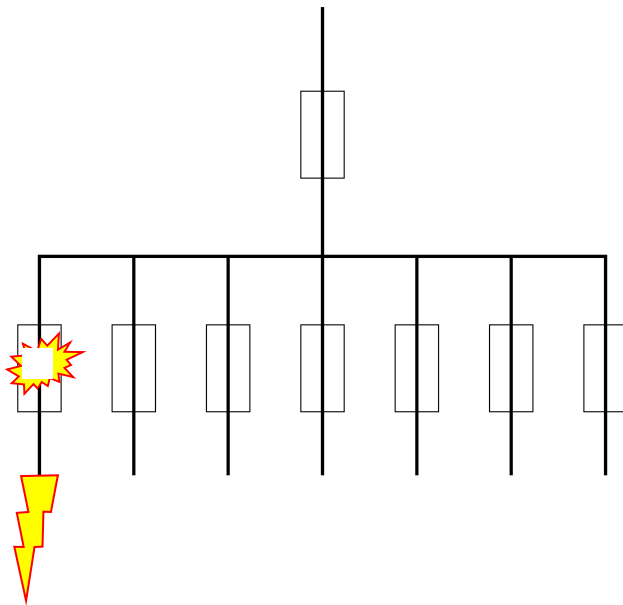
A fault in one circuit should not affect other circuits



AS/NZS 3000:2007
Clause 2.5.6
Page 90

Discrimination

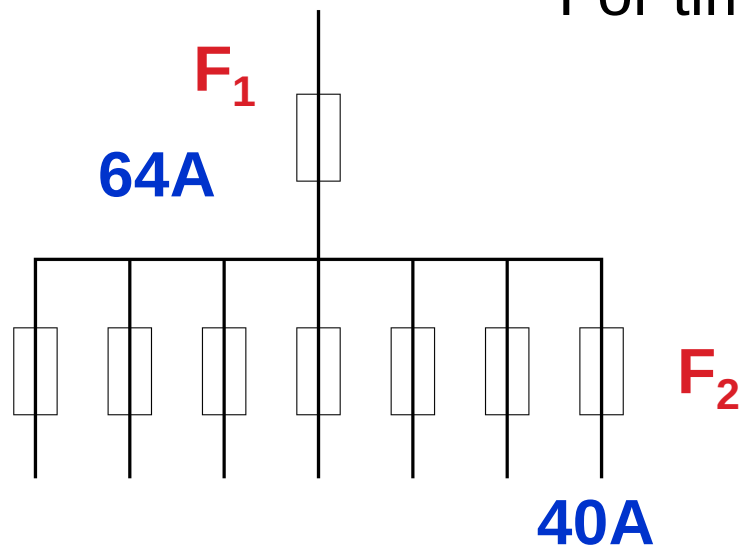
A fault in one circuit should not affect other circuits



Discrimination

A fault in one circuit should not affect other circuits

For times greater than 0.01 seconds



$$F_1 = F_2 \times 1.6$$

Discrimination

A fault in one circuit should not affect other circuits

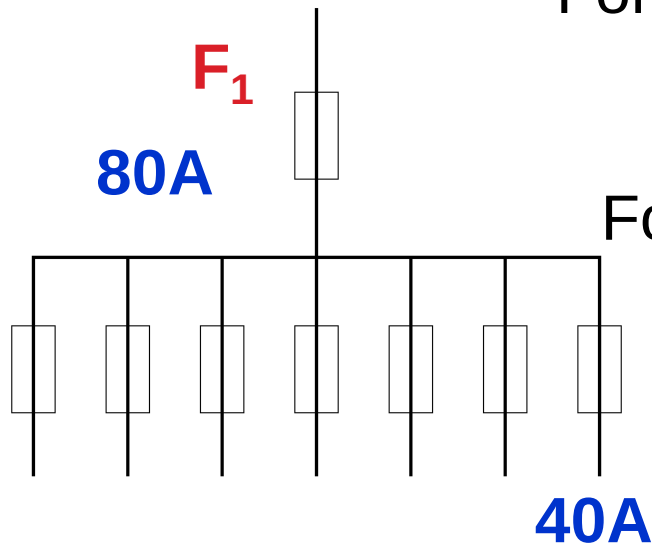
For times greater than 0.01 seconds

$$F_1 = F_2 \times 1.6$$

For times less than 0.01 seconds

$$F_1 = F_2 \times 2$$

$$(I^2t) F_1 = (I^2t) F_2 \times 2$$



$$F_1 = 2 \times F_2$$

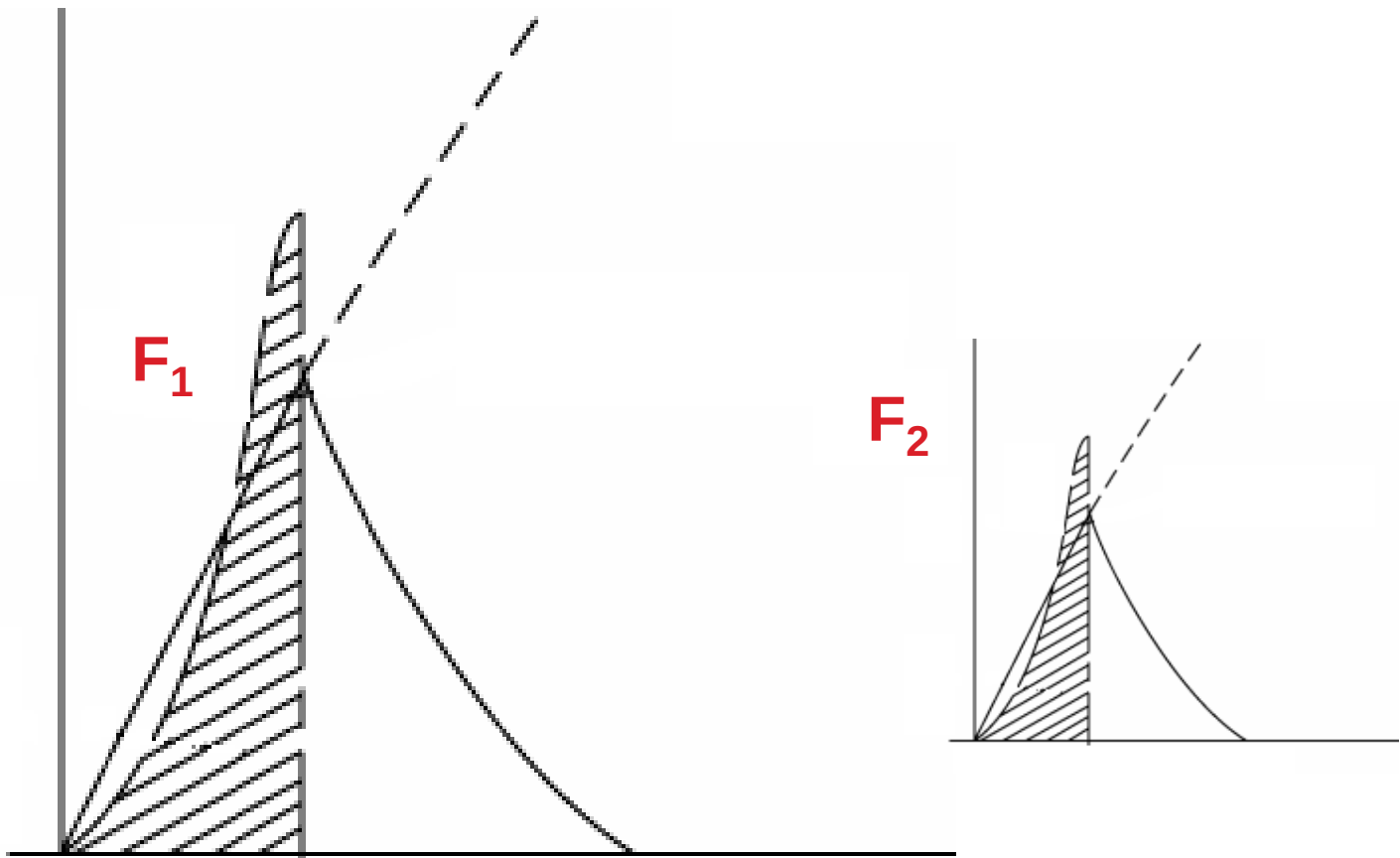


Figure 13.15(b) Time-current characteristic curves for 2 A to 800 A general fuse links

